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Examining energy inequality under the rapid residential energy transition in China through household surveys

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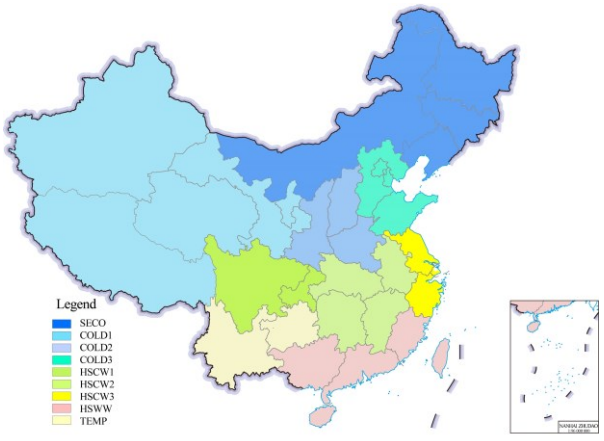
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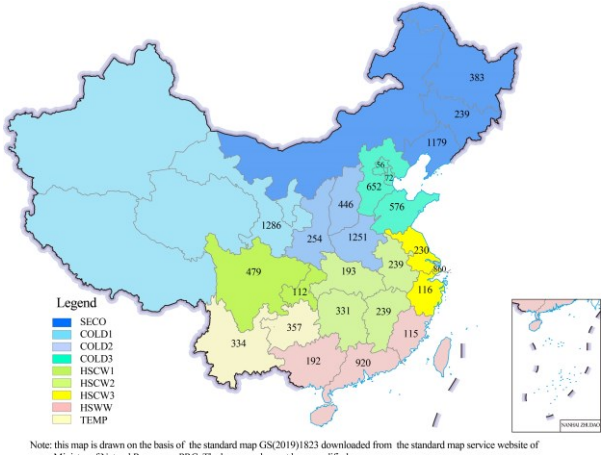
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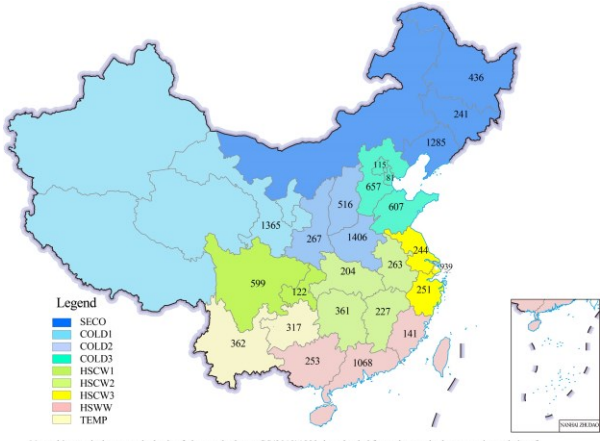
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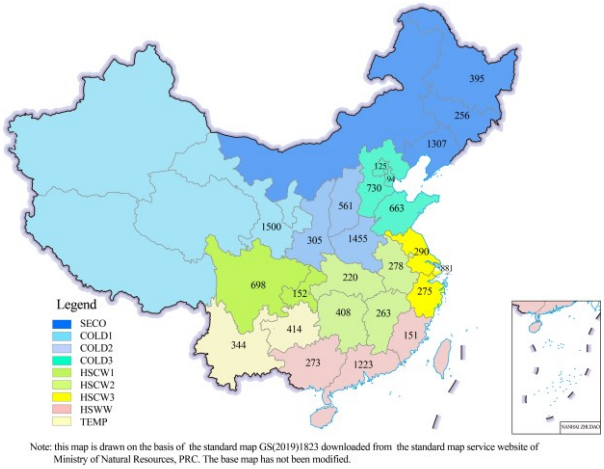
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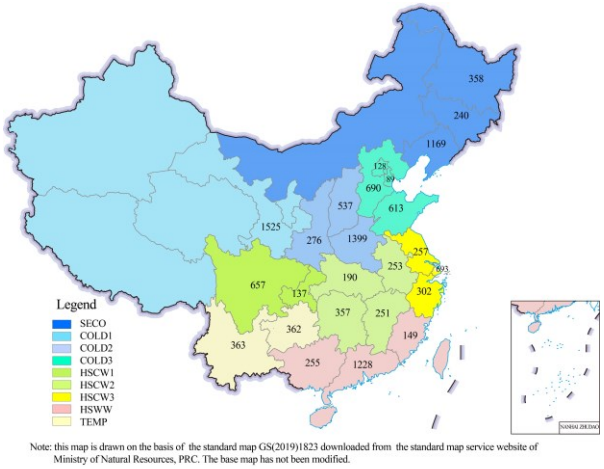
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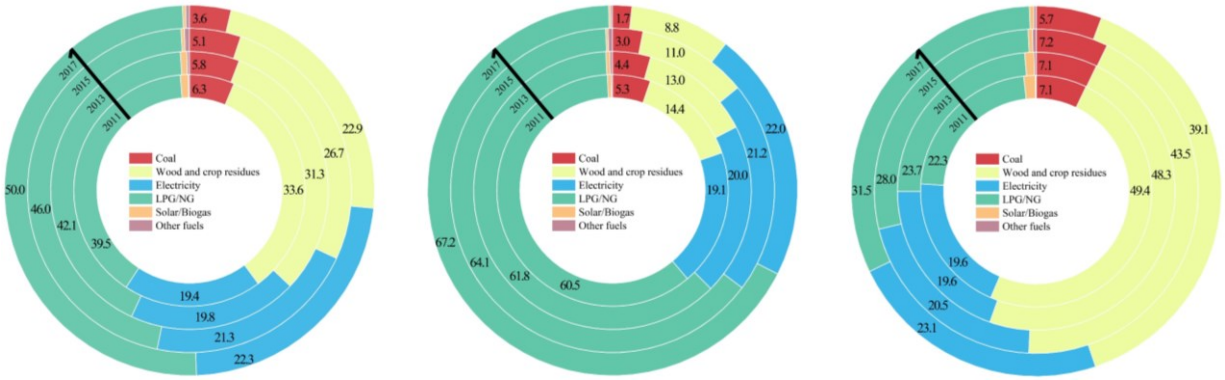


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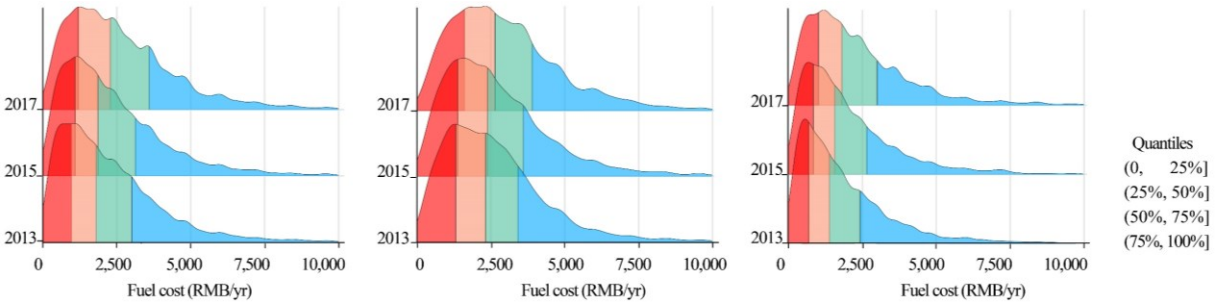
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Supplementary Figure 1 | Region definitions and sample distribution. **a**, classification of regions based on geographical proximity and economic development stage at the provincial scale[1]; **b**, regrouping **a** out of consideration for the regional differences in climate condition; **c-d,** distribution of sample size at the provincial level for CFPS 2012 (**c**), CFPS 2014(**d**), CFPS 2016(**e**), CFPS 2018(**f**).



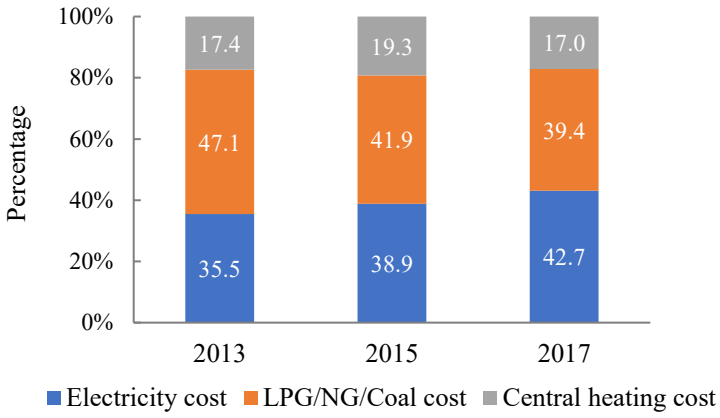
a **b** **c**

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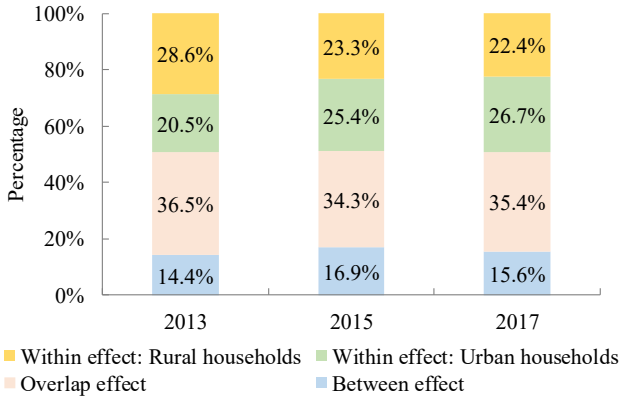


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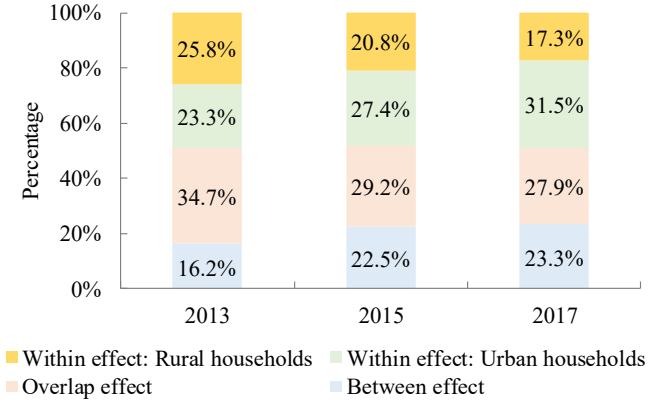
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a

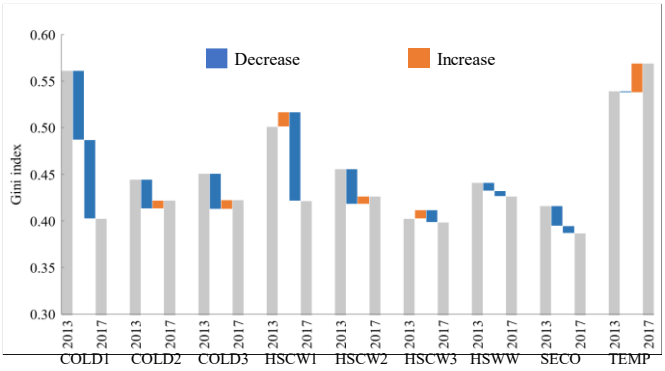


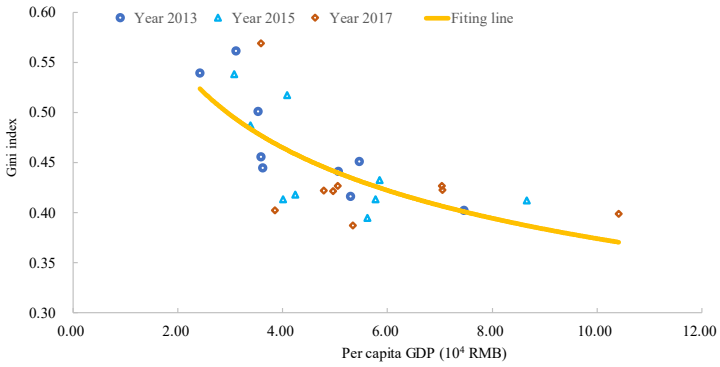
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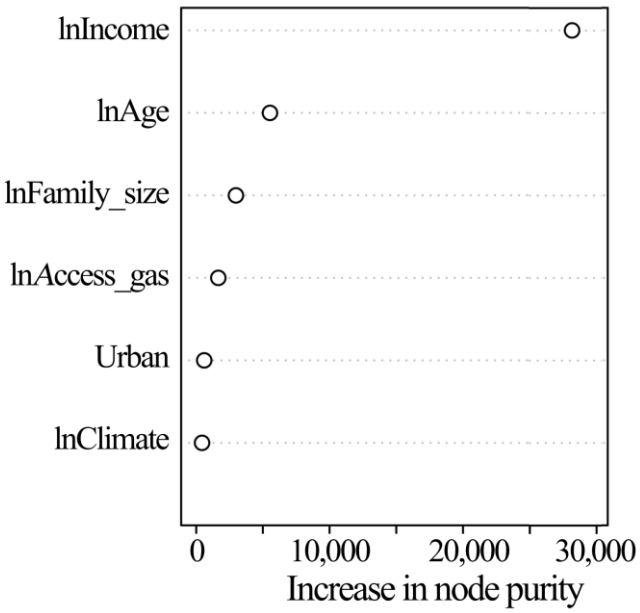
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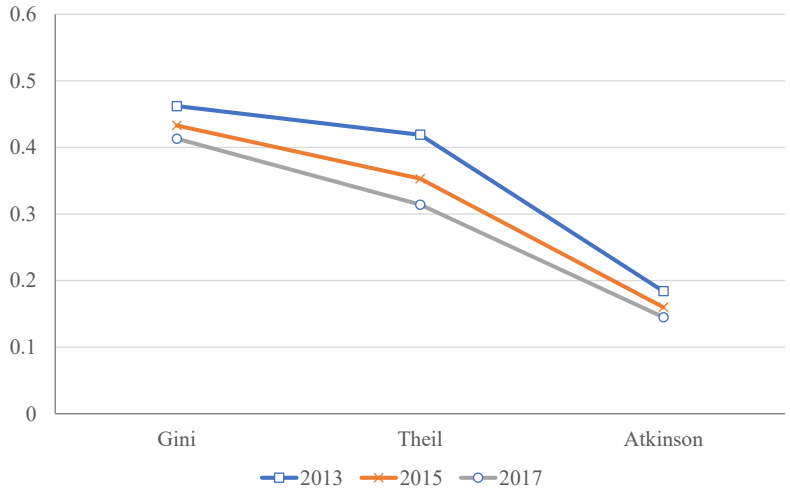


a **b**

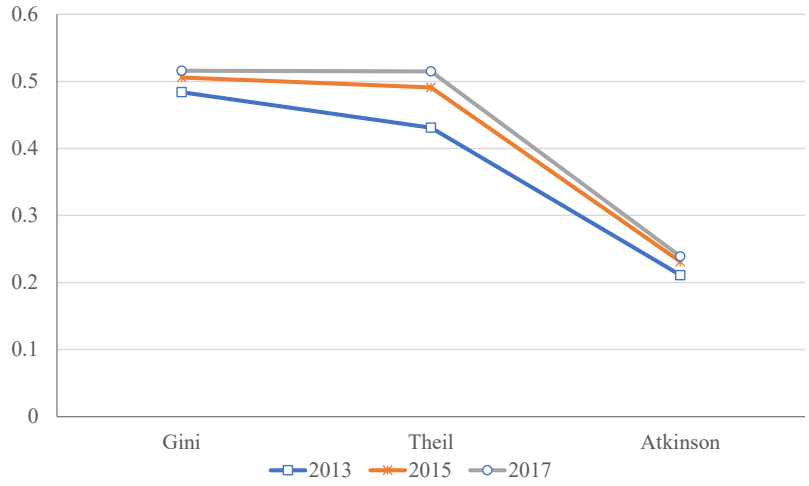
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Supplementary Figure 6 | Variable importance calculated by random forest regression model.



a



b

Supplementary Figure 7 | Comparison between Gini, Theil, and Atkinson index for fuel cost (a) and income (b). When employing the three indexes to quantify the household fuel cost and income inequalities, their patterns and trends highly agreed with each other, indicating the robustness of Gini index in this study.

Supplementary Tables

Supplementary Table 1 | Household income per capita across income groups in 2011, 2013, 2015, and 2017. (Unit: RMB). The income data used in this table were obtained from national statistic yearbooks. The statistical yearbooks reported average income patterns for different income groups based on the sampling survey conducted by the National Bureau of Statistics of China, and the entire households were split into 12 income groups (7 urban and 5 rural) according to the household numbers. The seven urban income groups were extremely poor (10% of the urban household number), poor (10%), lower middle (20%), middle (20%), middle high (20%), rich (10%), and very rich (10%). The five rural income groups were poor (20% of the rural household number), lower middle (20%), middle (20%), middle high (20%), and rich (20%). In this study, following the protocol by the statistic yearbooks, the urban lower middle- and middle households were grouped into the middle class. Finally, we classified all sample households into different income groups according to their average incomes for urban and rural separately.

Income group	Urban households				Rural households			
	2011	2013	2015	2017	2011	2013	2015	2017
Extremely poor	≤8788.9	≤9895.9	≤12230.9	≤13723.1	≤2000.5	≤2887.9	≤3085.6	≤3301.9
Poor	(8788.9, 14498.3]	(9895.9, 17628.1]	(12230.9, 21446.2]	(13723.1, 24550.1]	(2000.5, 4255.7]	(2887.9, 5965.6]	(3085.6, 7220.9]	(3301.9, 8348.6]
Middle	(14498.3, 19544.9]	(17628.1, 24172.9]	(21446.2, 29105.2]	(24550.1, 33781.3]	(4255.7, 6207.7]	(5965.6, 8438.3]	(7220.9, 10310.6]	(8348.6, 11978.0]
Middle high	(19544.9, 26420.0]	(24172.9, 32613.8]	(29105.2, 38572.4]	(33781.3, 38572.4]	(6207.7, 8893.6]	(8438.3, 11816.0]	(10310.6, 14537.3]	(11978.0, 16943.6]
High	(26420.0, 47021.0]	(32613.8, 57762.1]	(38572.4, 65082.2]	(38572.4, 77097.2]	(8893.6, 16783.1]	(11816.0, 21323.7]	(14537.3, 26013.9]	(16943.6, 31299.3]
Rich	>47021.0	>57762.1	>65082.2	>77097.2	>16783.1	>21323.7	>26013.9	>31299.3

Supplementary Table 2 | Median household energy cost by income groups in 2013, 2015, and 2017. (Unit: Ren Min Bi, Chinese currency, 1 Ren Min Bi≈0.16 US dollar)**.**

Income group	Urban households			Rural households		
	2013	2015	2017	2013	2015	2017
Extremely poor	2032	2040	2400	1180	1200	1440
Poor	2520	2400	3000	1200	1392	1800
Middle	2460	2640	2880	1380	1800	2000
Middle high	2483	2640	2900	1560	1800	2160
High	2400	2400	2760	1680	1800	2160
Rich	2400	2280	2480	1800	1960	2094

Supplementary Table 3 | Regression results with province-year fixed-effect model.

	Coefficients	Definition
lnIncome	-0.097** (-5.611)	The median of household annual income per capita in a province.
lnClimate	0.069 (0.482)	The sum of heating degree days and cooling degree days for a province.
Pol_intensity	-0.092** (-1.531)	A dummy variable distinguishing the difference in energy transition policy across the regions during 2013-2017.
Constant	1.116*** (3.265)	
Year	Y	
Province	Y	
Observations	75	
R-squared	0.490	
F	16.85 ***	

Note: *t* statistics are in parentheses; $*p < 0.1$, $**p < 0.05$, $***p < 0.01$. *p*-values are for a two-sided test based on normal distribution. Y denotes that the variable is controlled in the model.

Supplementary Table 4 | Region definition.

Climate zone	Region	Provinces/municipalities included in each region
Severe cold winter (SECO)	Northeast (SECO)	Liaoning, Jilin, Heilongjiang, and Inner Mongolia
Cold winter (COLD)	Northwest (COLD1)	Gansu, Qinghai, Ningxia, and Xinjiang
	Central (COLD2)	Shannxi, Shanxi, and Henan
	North (COLD3)	Beijing, Tianjin, Hehei, and Shandong
Hot summer and cold winter (HSCW)	Southwest (HSCW1)	Sichuan, Chongqing
	Central (HSCW2)	Anhui, Hubei, Hunan and Jiangxi
	Yangtze River Delta (HSCW3)	Shanghai, Jiangsu and Zhejiang
Hot summer and warm winter (HSWW)	Southeast (HSWW)	Guangxi, Hainan, Guangdong, and Fujian
Temperate weather through all year (TEMP)	Southwest (TEMP)	Yunnan and Guizhou

Supplementary Table 5 | Decomposition of Gini coefficient by region.

Region	In 2013	In 2015	In 2017
COLD1	2.0%	2.2%	2.0%
COLD2	2.6%	1.9%	2.1%
COLD3	1.3%	1.5%	1.7%
HSCW1	0.3%	0.4%	0.3%
HSCW2	0.6%	0.6%	0.7%
HSCW3	1.2%	1.1%	1.0%
HSWW	1.4%	1.5%	1.9%
SECO	2.5%	2.3%	2.1%
TEMP	0.2%	0.3%	0.3%
Between effect	21.6%	16.2%	16.5%
Overlap effect	66.3%	71.9%	71.5%

Supplementary Table 6 | Distribution of median energy burden by urban-rural settings and regions during 2013-2017.

Region	Year 2013 (%)			Year 2017 (%)			Difference（2017-2013）(%)		
	Whole	urban	rural	Whole	Urban	Rural	Whole	Urban	Rural
COLD1	6.88	7.93	6.52	7.6	6.65	7.95	0.72	-1.28	1.43
COLD2	5.22	5.87	4.7	6.59	5.72	7.43	1.37	-0.15	2.73
COLD3	6.8	6.65	7.05	7.23	5.74	9.25	0.43	-0.91	2.2
HSCW1	6.67	5.29	6.56	6.34	4.71	7.61	-0.33	-0.58	1.05
HSCW2	4.49	4.8	3.59	4.62	4.69	4.5	0.13	-0.11	0.91
HSCW3	2.94	2.86	3.18	3.32	2.83	4.34	0.38	-0.03	1.16
HSWW	5.51	5.87	4.93	5.49	5.1	5.84	-0.02	-0.77	0.91
SECO	6.81	7.16	6	7.5	6.68	9.33	0.69	-0.48	3.33
TEMP	4.12	4.51	3.73	4.14	3.09	4.43	0.02	-1.42	0.7

Supplementary Table 7 | Test in the correlation of instrumental variables with the core explanatory viable.

Variables	ln*Inc_per capita*
lnShare_off_farm	0.053**
	(8.45)
lnAge	0.197***
	(2.87)
lnFamily_size	-0.110***
	(-4.69)
Urban	0.101**
	(2.02)
lnAccess_gas	-0.053*
	(-1.73)
lnClimate	-0.013
	(-0.32)
Constant	8.517***
	(29.50)
Year	Y
Household	Y
Observations	34,992
R-squared	0.521
F	24.81***
Critical values for weak-identification test	
10% maximal IV size	16.38
15% maximal IV size	8.96
20% maximal IV size	6.66
25% maximal IV size	5.53

Note: *t* statistics are in parentheses; $*p < 0.1$, $**p < 0.05$, $***p < 0.01$. *p*-values are for a two-sided test based on normal distribution. Y denotes that the variable is controlled in the model.

Supplementary Table 8 | Target sample sizes of CFPSs by regions and their response rates.

Type of provinces	Provinces or equivalents (municipalities/autonomous regions)	Target Number of Households	Response rate in CFPS 2012	Response rate in CFPS 2014	Response rate in CFPS 2016	Response rate in CFPS 2018
Self-representative provinces	Shanghai	1,600	64.9%	63.2%	57.4%	53.2%
	Liaoning	1,600	86.1%	86.3%	85.1%	81.0%
	Henan	1,600	91.6%	96.5%	94.9%	94.4%
	Gansu	1,600	93.4%	93.1%	95.4%	99.9%
	Guangdong	1,600	74.2%	80.8%	83.3%	91.4%
Non-self-representative	Jiangsu, Zhejiang, Fujian, Jiangxi, Anhui, Shandong, Hebei, Shanxi, Jilin, Heilongjiang, Guangxi, Hubei, Hunan, Sichuan, Guizhou, Yunnan, Tianjin, Beijing, Chongqing, Shaanxi	8,000	84.4%	90.3%	92.0%	94.0%
The whole sample		16,000	83.2%	87.2%	87.6%	89.0%

Supplementary Table 9 | Profile of household characteristics.

Indicators	CFPS 2012	NBS 2012	CFPS 2014	NBS 2014	CFPS 2016	NBS 2016	CFPS 2018	NBS 2018
Number of permanent residents per household (Persons)	3.27	3.02	3.21	2.98	3.33	3.10	3.45	3.03
Male percentage (%)	50.5	51.3	50.4	51.3	50.6	51.3	50.0	51.2
Share of population aged 0-14 (%)	16.4	16.5	16.4	16.4	17.2	16.5	14.6	16.8
Share of population aged 15-64(%)	72.7	74.1	72.4	73.9	70.5	73.0	73.0	71.8
Share of population aged 65 or over (%)	10.9	9.1	11.2	9.7	12.4	10.5	12.4	11.4
Per capita income for urban household (RMB)	23365	23979	26676	29547	29415	31195	34943	36396
Per capita income for rural household (RMB)	9569	9833	10286	10991	10835	11422	13995	13432
Share of salary in urban household income (%)	59.3	64.3	62.2	64.1	62.6	62.0	63.2	61.0
Share of salary in rural household income (%)	38.3	34.3	38.6	31.4	39.0	40.3	43.2	40.9
Per capita cost for urban household (RMB)	19705	20366	18676	18023	19301	21392	22153	24446
Per capita cost for rural household (Yuan)	8168	8642	10286	9606	10902	9223	12137	10955

Note: The official data in the table comes from the China Statistical Yearbook 2012, 2014, 2016 and 2018, compiled by National Bureau of Statistics of China (NBS).

Supplementary Table 10 | Comparison between proportions of households reporting fuel use for cooking based on the tracking sample and the total sample.

Type of residential fuel	CFPS 2012		CFPS 2014		CFPS 2016		CFPS 2018	
	Tracking sample	Total sample	Tracking sample	Total sample	Tracking sample	Total sample	Tracking sample	Total sample
Coal	6.4%	6.3%	6.0%	5.8%	5.2%	5.1%	3.9%	3.6%
Wood and crop residues	34.6%	33.6%	32.7%	31.3%	28.4%	26.7%	24.8%	22.9%
Electricity	19.3%	19.4%	19.9%	19.8%	21.5%	21.3%	22.5%	22.9%
LPG/NG	38.4%	39.5%	40.3%	42.1%	44.0%	46.0%	48.3%	50.0%
Solar/Biogas	1.2%	1.1%	0.9%	0.8%	0.4%	0.3%	0.3%	0.3%
Other fuels	0.2%	0.2%	0.2%	0.3%	0.5%	0.5%	0.2%	0.3%

Supplementary Notes

Supplementary Note 1: Definitions of energy transition, energy cost, energy burden, energy poverty, and energy inequality

In this study, the term 'energy transition' was defined as the energy shift from the traditional solid fuels to gas fuels and electricity during our research period. Note that if a household has a convenient access to clean energy driven by energy transition policies and actions, but concurrently use both of traditional solid-and clean fuels, we also regarded the household undergoing energy transition because it also fairly represents a special group. If we excluded it from the whole samples, the household energy cost and its related energy burden are likely to be overestimated.

Energy cost in this study was measured by how much a household spends on energy bills per year, while 'energy burden' was defined as the percentage of household income spent on energy bills. Based on this definition, existing studies have developed a variety of thresholds to assess the magnitude of energy burden. For example, the United Kingdom (UK) has officially incorporated energy poverty formally into its government programs since 2000, applying an energy burden metric to assess whether households are facing energy poverty. This metric has evolved from a simple ratio of household income and energy expenditures (greater than 10%) to one that incorporates building efficiency ratings and average incomes in the community[2,3]. Meanwhile, similar metrics have been developed in member countries of the European Union (EU). In the US, 'energy poor' households commonly refer to those with an expenditure of greater than 6% of household income on meeting energy costs[4-7], given energy expenditures should not be greater than 20% of housing expenses, which themselves should not exceed 30% of household income[7]. Moreover, Cook and Shah (2018) distinguished between 'energy stressed' households with energy burdens of 4%–7%, 'energy burdened' households with 7%–10% energy burdens, and 'energy impoverished' households with energy burdens greater than 10%[8]. In short, there lacks a universal and unified threshold or criterion for energy poverty assessment, however, the threshold of 10% is relatively more common across the world[9]. Therefore, here we categorize households as living in 'energy poverty' when they spend more than 10% of their household income on energy costs.

The concept of 'Energy inequality' has been defined from a multidimensional perspective in previous literature[4,10-13]. In this study, we defined it from two perspectives, energy cost-based inequality and energy burden-based inequality. The former can be quantitively estimated using Gini index, while the latter was discussed and analyzed by comparison between urban and rural households across different regions or income-groups.

Supplementary Note 2: Energy transition actions in China from 2013 to 2017

With the implementation of the toughest-ever clean air policy in China, Air Pollution Prevention and Control Action Plan (APCAP) [14], a new campaign that substitute residential solid fuels (coal, biomass) with clean fuels such as gas and electricity has been launched by China's central and local governments since 2013. In this regard, the

year 2013 also featured a milestone: the residential coal-switch policy's transformation from a local pilot program to a large-scale regional application. These measures mainly include decreasing the penetrations of direct coal-burning by replacing coal with natural gas and electricity, with support of subsidizing household natural gas consumption and constructing necessary infrastructures. Meanwhile, China's central government proposed an ambitious goal for Beijing, Tianjin, and 26 other municipalities in the North, namely the clean heating rate should increase to 50% by 2019 and 70% by 2021, in particular, it was planned that 60% of rural coal-using households will shift to clean fuels by 2021[15]. Along with the full implication of these measures, by the end of 2017, energy consumption in 6 million households in China switched from coal to electricity and natural gas[16].

In addition, as the socioeconomic development, China has also witnessed a rapid energy transition in residential sector during 2011-2017[16,17]. Here, based on the valid data from 11,134-household survey in 2012, 12,946-household survey in 2014, 13,261-household survey in 2016, and 12,418-household survey in 2018, we found that the share of households using wood and crop residues decreased from 33.6%% in 2011 to 31.3% in 2013. Since 2013, the strict implementation of APCAP has led to a considerable drop in the share of households using wood and crop residues, from 31.3% to 26.7% in 2015, and 22.9% in 2017. Meanwhile, although the share of households using gas increased obviously during 2011-2013, the growth rate became higher during 2013-2015, and 2015-2017, with the wide implementation of energy transition policy. In addition, as we observed, the immediate effect of energy transition policy was more significant in rural areas. These findings indicate the massive impacts of the APCAP and other energy transition measures on the residential energy pattern mix in the short term.

Supplementary Note 3: Evaluating the robustness of Gini index by comparing with Theil and Atkinson coefficients

Among a variety of inequality indices, the Lorenz curve and the Gini index are one of the most widely used analytical tools in the economics literature. There have been several studies to measure inequalities in the energy-related fields [18-20]. To assess the robustness of the Gini index, this study also applied the Theil and Atkinson indices to quantify the energy cost-based inequality trends. As shown in Supplementary Figure 7, when employing the three indexes to quantify the household fuel cost and income inequalities, their patterns and trends highly agreed with each other, indicating the robustness of the Gini index in this study.

Supplementary Note 4: Effect assessment of rapid energy transition on energy cost-based inequality at regional level

Considering the missing of specific household's address at the city-level due to privacy concerns, province-year fixed-effect model was employed to reveal the effect of rapid energy transition on provincial energy inequality. Here, we used $\ln G_{it}$ (representing the Gini index for i province in t year) as the dependent variable, and a dummy variable Pol_intensity was introduced to represent the energy transition campaign intensity at

the provincial level.

The Air Pollution Prevention and Control Action Plan was promulgated at the end of 2013 by the State Council of the People's Republic of China. Considering that it would take some time to further come up with plans and implement detailed actions for local governments, we assumed that the energy transition policies had a limited effect on the household's energy transition in 2013, thus all regions were assigned 0 in 2013. Meanwhile, according to the Air Pollution Prevention and Control Action Plan, Beijing-Tianjin-Hebei, Yangtze River Delta (including Shanghai, Jiangsu, Zhejiang), and Pearl River Delta (Guangdong) were regarded as the major targeted regions to replace coal with electricity, natural gas supply and the use of non-fossil energy. Thus, we assigned 1 to these seven cities or provinces, while 0 to the other provinces in 2015. In February 2017, the Clean Winter Heating Plan in Northern China was initiated and implemented to accelerate the adoption of the clean energy for heating using in the Northern China. Besides Beijing-Tianjin-Hebei, this plan also covered Shanxi, Shandong, and Henan. Thus, we assigned 1 to these ten cities or provinces in 2017, including Beijing-Tianjin-Hebei, Yangtze River Delta (including Shanghai, Jiangsu, Zhejiang), Pearl River Delta (Guangdong), Shanxi, Shandong, and Henan, and the remaining fifteen provinces were assigned 0. In addition, by referring to a study of Tao et al., (2019), two indicators associated with households' income (Income) and regional climate conditions (Climate) were also introduced into the model in their logarithmic form. The definitions of these independent variables and regression results are presented in Supplementary Table 3.

The results show that both indicators of energy transition policy intensity (Pol_intensity) and household income (lnIncome) were negatively associated with the Gini index, indicating that rapid energy transition under government's push will reduce energy inequality to some extent, as well as growth in household income. Meanwhile, this finding also caused a concern for the affordability issue for low-income households during this rapid energy switch from free or low-price solid fuels to relatively high-price clean fuel. In this regard, we think that it is necessary to further discuss the changes in energy burden and its driving factors during this rapid energy transition period.

Supplementary Note 5: Connecting energy burden to potential driving factors

Theoretically, drivers of high household energy burden could be roughly divided into the following five categories: household socioeconomic situation, housing characteristics, energy prices and policies, location and geographical conditions, and behavioral factors[21,22,23]. According to previous studies, considering the availability of data, this study evaluated a comprehensive set of household-level factors (household demographic characteristics and socioeconomic status) and a number of distal factors (gas access penetration, urban-rural settings, and climate conditions) that may be closely associated with energy burden.

As shown in Table 1 in the main text, Household-level variables include household income per capita (Income), the number of household members that live together (Family_size), the average age of family members (Age). In addition, a dummy variable Urban, referring to household's living conditions, was also considered into our models,

in order to evaluate the impact of urbanization on household energy burden. Regional-level variables include the market penetration of gas fuels (Access_gas) and climate conditions (Climate). Specifically, the value of Access_gas for each region was set by the provincial statistical data, which is evaluated at the urban and rural scales, respectively. In addition, the sum of Heating Degree Days (HHD) and Cooling Degree Days (CDD) for each province was calculated to serve as the value for Climate, which can be defined as follows[24,25,26]:

$$Climate = \frac{1}{n}\sum_{k=1}^{n}\sum_{i=1}^{I_t}\alpha(T_b - T_{kit}) + \frac{1}{n}\sum_{k=1}^{n}\sum_{i=1}^{I_t}(1-\alpha)(T_b - T_{kit}) \quad (1)$$

where n is the number of meteorological observation stations in a province, and I_t is the number of days in the t month; T_{kit} is defined as the daily average temperature in the k meteorological observation station on day i of month t. T_b denotes the reference temperature. In this study, we set T_b as 65°F by referring to previous studies[11,14]. α is a characteristic function, which is 1 when T_{kit} is more than T_b, otherwise, it is 0. The daily average temperature data were obtained from 665 meteorological observation stations in 2013, 648 meteorological observation stations in 2015, and 563 meteorological observation stations in 2017, which can be downloaded from http://data.sheshiyuanyi.com/WeatherData/.

Here, we constructed linear and non-linear models to understand which variables, if any, may correlate to household energy burden. To eliminate bias from unobservable, including those change over time but are constant over entities, and differences across households but are constant over provinces and time, the Ordinary Least Squares (OLS) regression models with household fixed-effects and province-year fixed-effects were estimated using unbalanced panel. Particularly, province-year fixed-effect model with clustered standard errors at provincial level was estimated considering that it can capture any other polices at provincial level that could influence the coefficients estimated in the empirical analyses.

Meanwhile, to control the endogeneity, this study incorporated an instrumental variable (IV), Share_off_farm to the linear model. As shown in Table 1, the indicator Share_off_farm indicates the share of laborers engaged in nonagricultural employment in a family. One of reasons for choosing this indicator lies in that we observed that 91.5% of rural households derived some income from off-farm sources. On average, off-farm income contributed more than 85% of total income in these households. In this regard, the share of laborers engaged in nonagricultural employment is likely to closely relate with household income level, which is unlikely to be affected by individual household energy burden. To confirm the validity of IV, we conducted correlation test for it, the results were showed in the Supplementary Table 7. It can be found that this IV correlated with the core explanatory variable (Income). Finally, the under-identification and weak-identification tests for the IV was conducted, the results were showed in the Table 2 in the main text, and the critical values for weak-identification test were presented in Supplementary Table 7, which implied that the IV was valid.

Supplementary References:

1. The standard map service website of Ministry of Natural Resources of PRC. China's map (2019). http://bzdt.ch.mnr.gov.cn/browse.html?picId=%224o28b0625501ad13015501ad2bfc0256%22.
2. Bednar, D. J. & Reames, T. G. Recognition of and response to energy poverty in the United States. *Nat. Energy* **5**, 432–439 (2020).
3. Alkon, M., Harish, S. P. & Urpelainen, J. Household energy access and expenditure in developing countries: Evidence from India, 1987–2010. *Energy Sustain. Dev.* **35**, 25-34 (2016).
4. Drehobl, A., L. Ross, & R. Ayala. How High are Household Energy Burdens? Washington, DC: American Council for an Energy-Efficient Economy (ACEEE, 2020); https://www.aceee.org/research-report/u2006.
5. Malakar, Y. & Day, R. Differences in firewood users' and LPG users' perceived relationships between cooking fuels and women's multidimensional well-being in rural India. *Nat. Energy* **5**, 1022-1031 (2020).
6. Sovacool, B. K. & Dworkin, M. H. Energy justice: conceptual insights and practical applications. *Appl. Energy* **142**, 435–44li4 (2015).
7. Brown, M. A. et al. Low-income energy affordability in an era of U.S. energy abundance. *Prog. Energy* **2**,042003 (2020).
8. Cook, J.J. & Shah, M. Reducing energy burden with solar: colorado's strategy and roadmap for states. *National Renewable Energy Laboratory*. https://www.nrel.gov/docs/fy18osti/70965.pdf (2018).
9. Ivanova, D. & Middlemiss, L. Characterizing the energy use of disabled people in the European union towards inclusion in the energy transition. *Nat. Energy* **6**, 1188–1197 (2021).
10. Wang, Q., Kwan, M.P., Fan, J. & Lin, J. Racial disparities in energy poverty in the United States. *Renew. Sustain. Energy Rev*. **137**, 110620 (2021).
11. Hernandez, D. et al. Energy Burden and the Need for Integrated Low-Income Housing and Energy Policy. *Poverty & Public Policy* **2**, 5-25 (2010).
12. Liddell, C. et al. Measuring and monitoring fuel poverty in the UK: National and regional perspectives. *Energy Policy* **49**, 27-32 (2012).
13. Lin, B. et al. Does off-farm work reduce energy poverty? Evidence from rural china. *Sustainable Production and Consumption* **27**, 1822-1829 (2021).
14. State Council of the People's Republic of China, Notice of the general office of the state council 591 on issuing the air pollution prevention and control action plan. http:// 592 www.gov.cn/zwgk/2013-09/12/content_2486773.htm. Accessed 21 August 2020.
15. National Development and Reform Commission, Clean Winter Heating Plan in Northern China (2017-2021) (2017). http://www.ndrc.gov.cn/zcfb/zcfbtz/201712/t20171220_871052.html. Accessed 27 March 2021.
16. Zhang, Q. et al. Drivers of improved PM2.5 air quality in China from 2013 to 2017. *PNAS* **116**, 608 24463–24469 (2019).
17. General Office of the Ministry of Housing and Urban-Rural Development. Notice of the general office of the ministry of housing and urban-rural development on printing and distributing the technical guidelines for rural pipeline natural gas engineering (2018). http://www.sohu.com/a/276915360_652961. Accessed 27 March 2021.
18. Wu, S., Zheng, X. & Wei, C. Measurement of inequality using household energy consumption data in rural China. *Nat. Energy* **2**, 795-803 (2017).

19. Wang, Q., Kwan, M.-P., Fan, J. & Lin, J. Racial disparities in energy poverty in the United States. *Renew. Sustain. Energy Rev*. **137**, 110620 (2021).
20. Kammen, D. M. & Kirubi, C. Poverty, energy, and resource use in developing countries. *Ann. NY Acad. Sci* **1136**, 348–357 (2008).
21. Drehobl, A., L. Ross, & R. Ayala. How High are Household Energy Burdens? Washington, DC: American Council for an Energy-Efficient Economy (ACEEE, 2020); https://www.aceee.org/research-report/u2006.
22. Sovacool, B. K. & Dworkin, M. H. Energy justice: conceptual insights and practical applications. *Appl. Energy* **142**, 435–444 (2015).
23. Brown, M. A. et al. Low-income energy affordability in an era of U.S. energy abundance. *Prog. Energy* **2**,042003 (2020).
24. Ahmed, T., Muttaqi, K.M. & Agalgaonkar, A.P. Climate change impacts on electricity demand in the State of New South Wales. *Applied Energy* **98,** 376-383 (2012).
25. Fan, J.L. et al. Impact of climatic factors on monthly electricity consumption of China's sectors. *Natural Hazards* **75**, 2027-2037 (2015).
26. Sivak,M. Potential energy demand for cooling in the largest metropolitan areas of the world: Implications for developing countries. *Energ. Policy* **37(4)**, 1382-1384 (2009).